

9/18/2026

Becoming a Data Head

CHAPTERS 13 & 14

WHEN NUMBERS MISLEAD

Pitfalls, Biases & Communication

Presented by Shayan Najafian

THE BIG IDEA

THE BIG IDEA

Two ways data projects fail

The data fools us

or

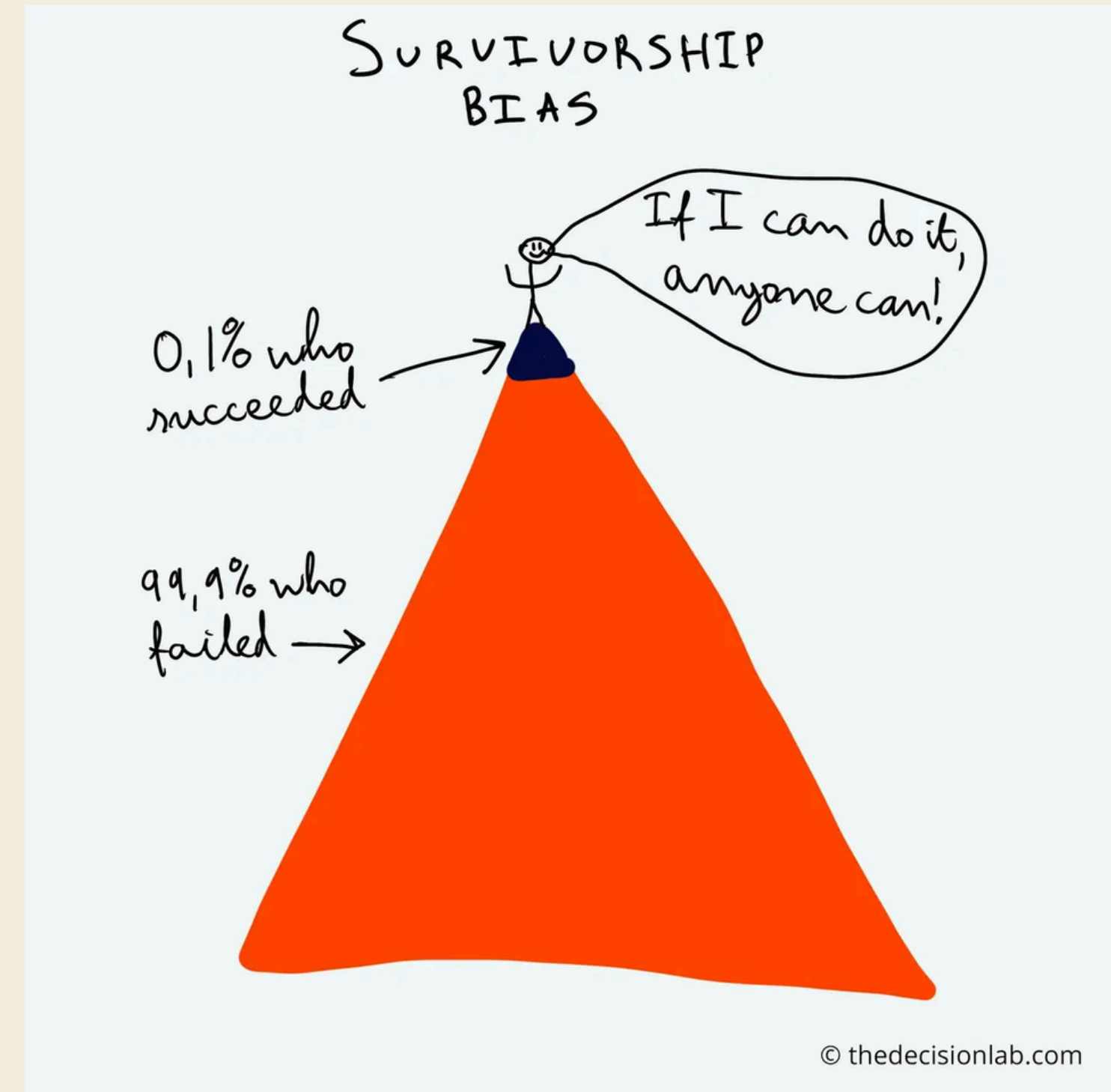
We fool ourselves

Real data can still produce the wrong story.



SURVIVORSHIP BIAS

We study what survived.
But what about:
what disappeared?

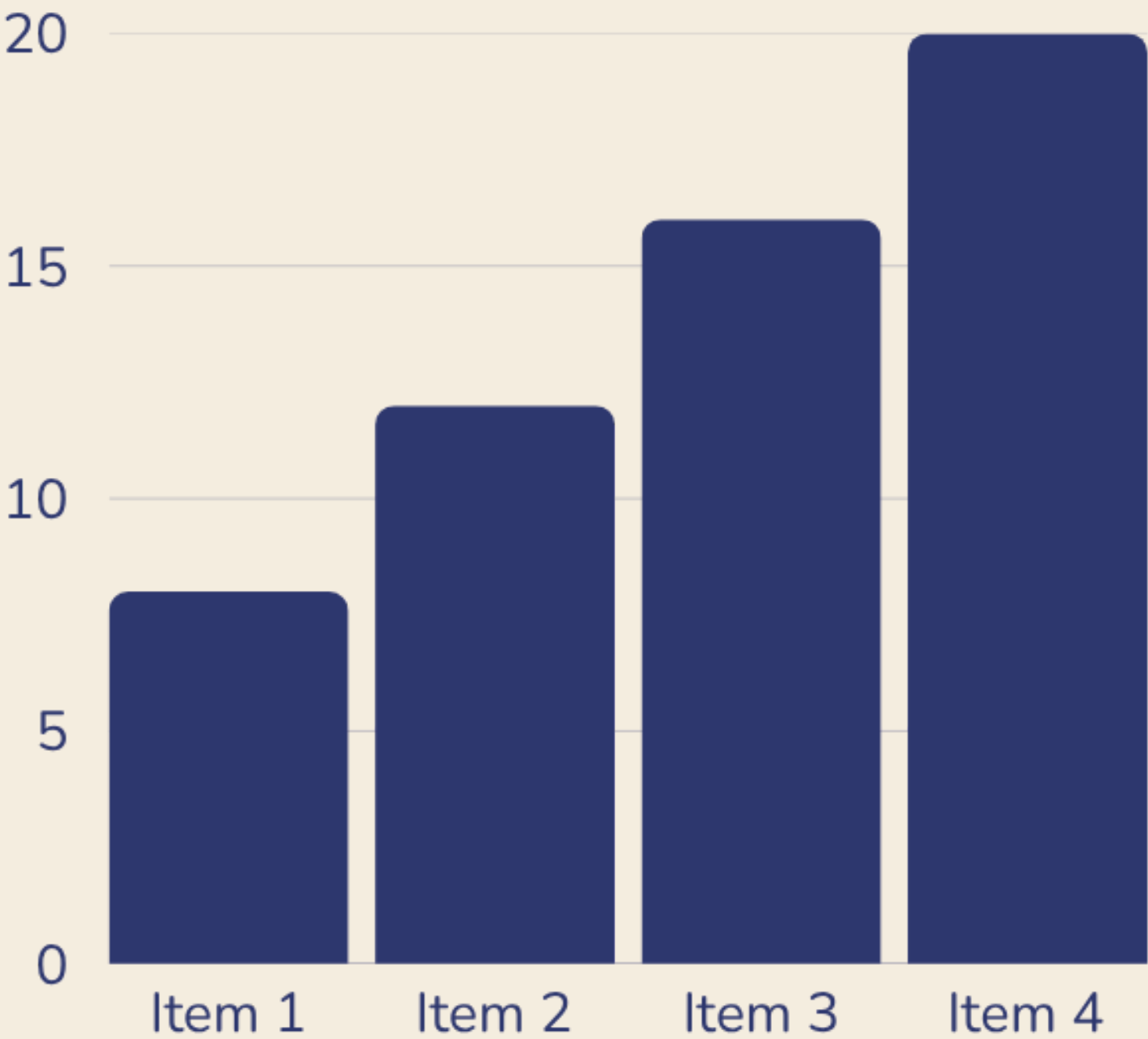


SURVIVORSHIP BIAS

ONLY SHOW THE WINNERS

- 50 funds launched
- Bad funds closed
- Only survivors remain
- Performance looks amazing

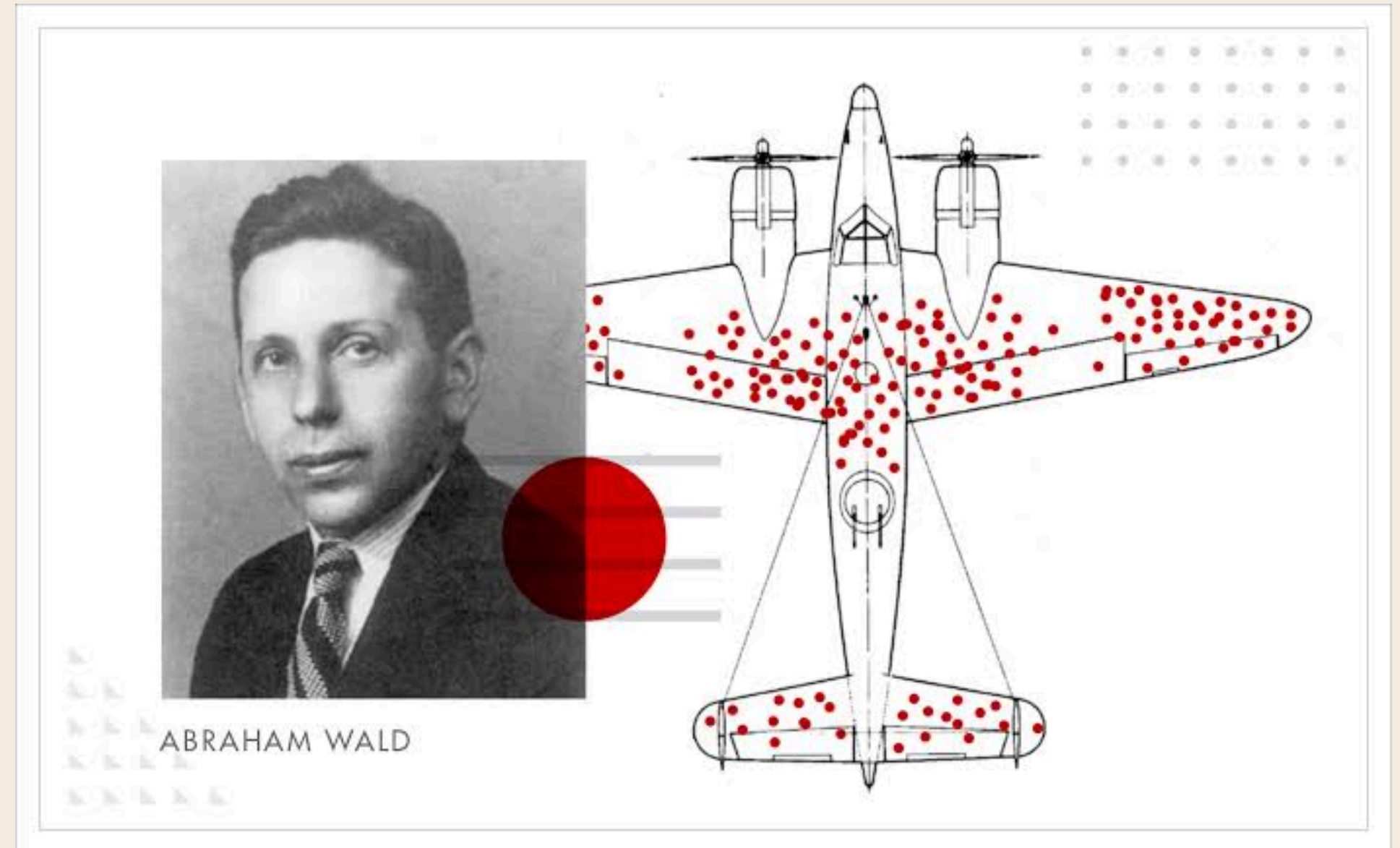
Ask: what failed and disappeared?



SURVIVORSHIP BIAS

WHERE WOULD YOU ADD ARMOR?

The visible bullet holes are from planes that came back.



Sometimes the missing cases matter more than the visible ones.

DATA HEAD QUESTION

SURVIVORSHIP BIAS

Who is missing from this dataset?

REGRESSION TO THE MEAN

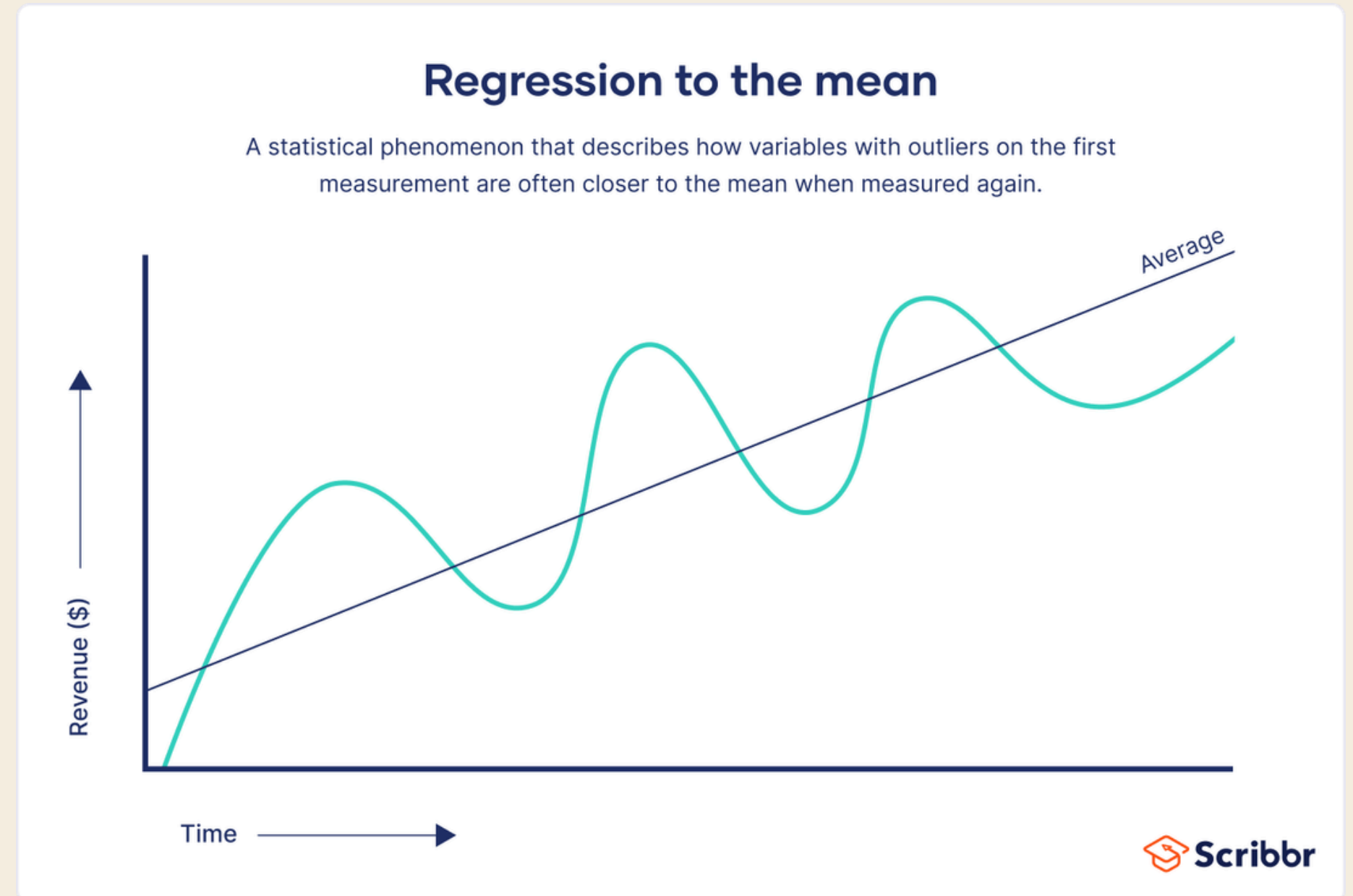
EXTREME TODAY MORE NORMAL TOMORROW

Outliers often move back toward ordinary values.

Not because of a curse.

Often because luck does not repeat.

Do not build a big story around one extreme point.



REGRESSION TO THE MEAN

PERFORMANCE = ABILITY + NOISE

Observed result

What we see in the data

Can be unusually high

Or unusually low

Underlying reality

Skill, system, process

plus random variation

closer to the mean

REGRESSION TO THE MEAN

THE MADDEN CURSE

Amazing season

Madden cover

Worse next season

Curse? Or regression to the mean?



Extreme success is hard to repeat.

REGRESSION TO THE MEAN

DID THE INTERVENTION WORK?

What happened

Poor performance

Manager intervenes

Performance improves

What to ask

Was it really caused?

Or was the first year just extreme?

Where is the comparison group?

SIMPSON'S PARADOX

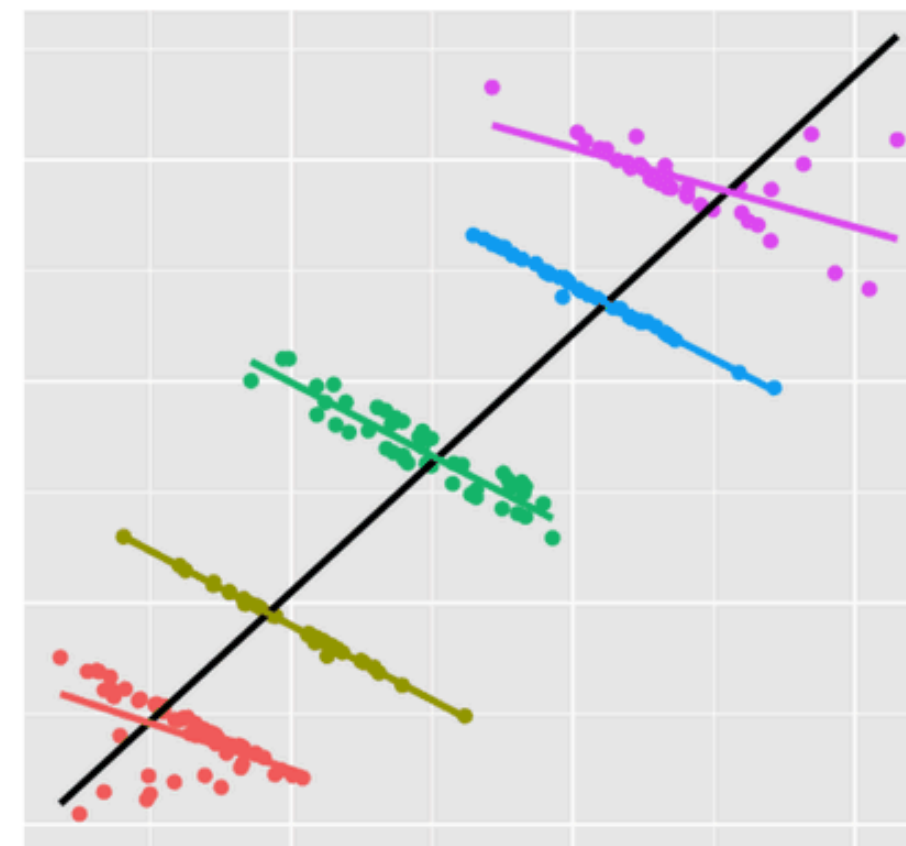
SIMPSON'S PARADOX

The overall trend says one thing.

The subgroups say the opposite.

A hidden variable changes the story.

Aggregates can hide the truth.



INTERACTIVE

WHICH TREATMENT IS BETTER?

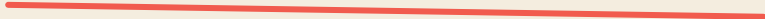


Treatment	Overall Success
Traditional	78%
New	83%

Ask the audience before revealing the split.

SIMPSON'S PARADOX

NOW SPLIT BY STONE SIZE



Treatment	Small Stones	Large Stones	Overall
Traditional	93%	73%	78%
New	87%	69%	83%

Traditional wins inside both groups, but loses overall.

SIMPSON'S PARADOX

HOW CAN BOTH BE TRUE?

New procedure

Mostly easier cases

More small stones

Overall rate looks better

Traditional procedure

More difficult cases

More large stones

Overall rate is pulled down

SIMPSON'S PARADOX

**What happens if we break the average into
meaningful groups?**

PROJECT PITFALL

EFFORT BIAS SUNK COST FALLACY

We already spent:

6 months

\$500,000

thousands of hours

We cannot stop now.

Past effort is not a reason to keep making a bad decision.



SUNK COST

THE CHECKPOINT QUESTION

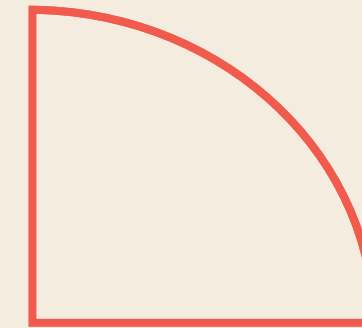
**Knowing what we know today, would we start
this project?**

P-HACKING

P-HACKING

Keep testing many relationships...
until one result becomes significant.
Then only report the survivor.

p=.41	p=.63
p=.72	p=.84
p=.19	p=.31
p=.048	p=.57



A small p -value is less impressive when many tests were tried.

P-HACKING

RUN ENOUGH TESTS...

Test 1 -> $p = .41$

Test 2 -> $p = .72$

Test 3 -> $p = .19$

Test 4 -> $p = .048$ <-- publish this

Test 5 -> $p = .63$

P-HACKING

P-HACKING IS ALSO SURVIVORSHIP BIAS

Many tests performed

Most are insignificant

They disappear

Nobody sees them

Only significant results survive

Looks like strong evidence

But may be chance

Ask how many tests ran

PRESSURE 2026

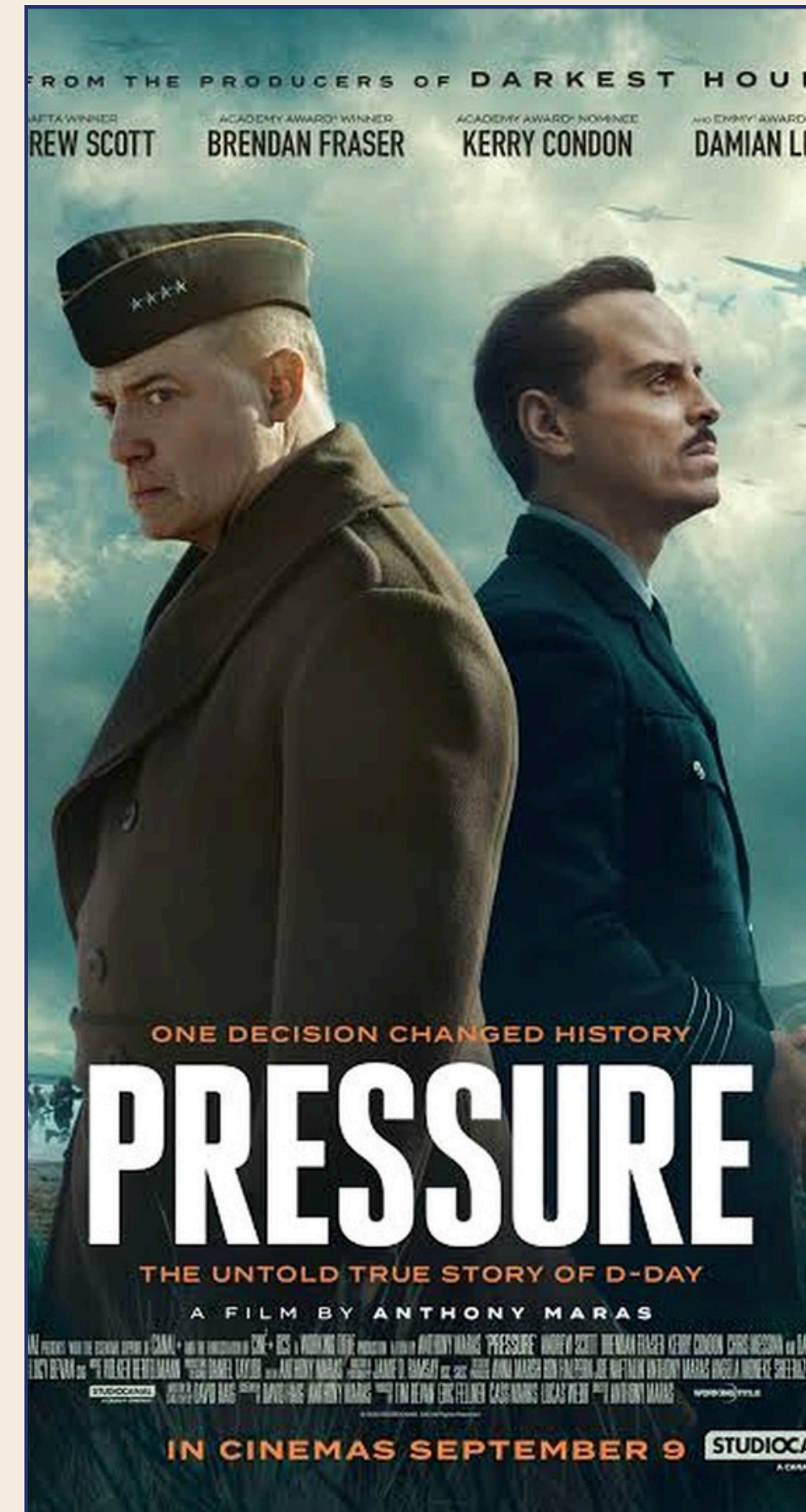
D-Day depended on an uncertain weather forecast.

Leaders wanted clarity.

The data did not give perfect certainty.

That pressure makes confirmation bias tempting.

Under pressure, people often prefer evidence that supports the decision they already want.



BIAS

CONFIRMATION BIAS

We search for evidence that confirms our beliefs.

We discount evidence that challenges us.

Data teams can become tools for justifying a decision already made.

A Data Head reports what is true, not what is convenient.

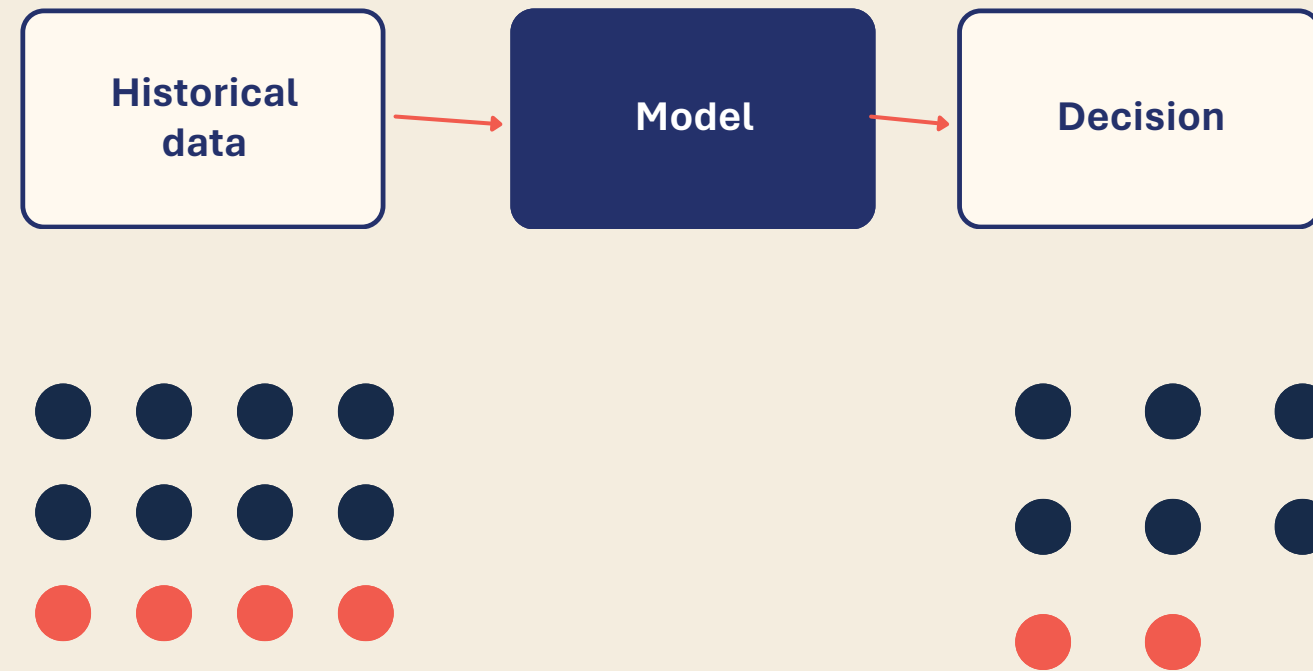


BIAS

ALGORITHMIC BIAS

Models learn patterns from history.
If history contains bias, the model can
reproduce it.

Automation can scale old mistakes.



Predictions are products of assumptions, not final truth.

SUMMARY

FIVE PITFALLS, ONE PATTERN

Pitfall	What are we missing?
Survivorship Bias	Missing cases
Regression to the Mean	Random variation
Simpson's Paradox	Hidden groups
Sunk Cost Fallacy	Future vs. past
p-hacking	Failed tests

We see only part of the story, then build a convincing explanation.

SUMMARY

Best sources to learn cognitive biases

Thinking , fast and slow by Daniel Kahneman
Azarakhsh Mokri: Youtube

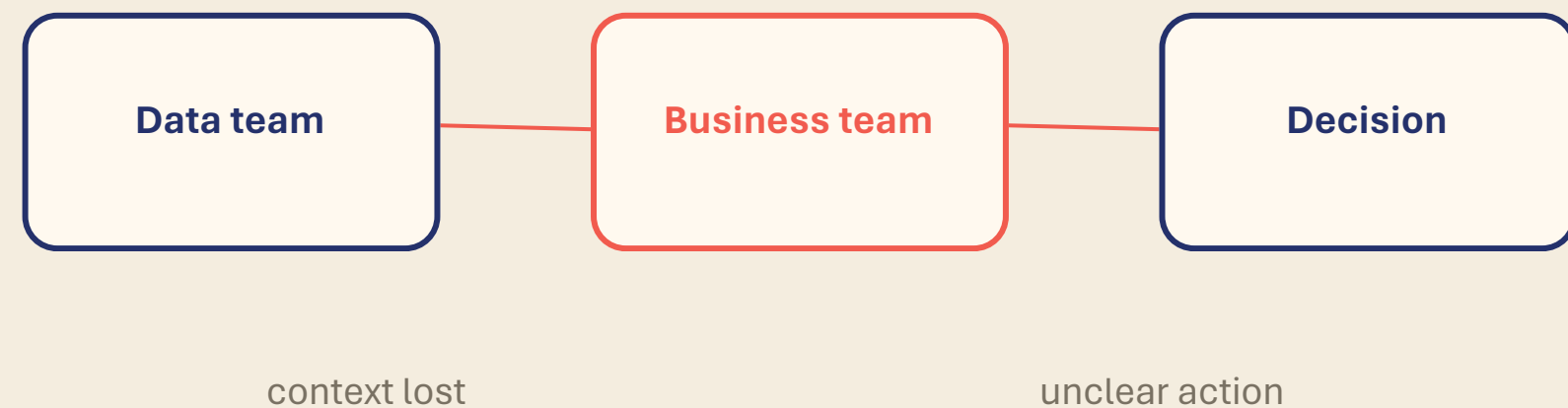
CHAPTER 14

GOOD MODELS CAN STILL FAIL

The math can be right.

The message can still fail.

Communication can destroy the value of good analysis.



Chapter 14 is about people, not algorithms.

COMMUNICATION BREAKDOWNS

SEVEN SCENES

Postmortem

expert called too late

Storytime

nuance removed

Telephone Game

statistic loses context

Into the Weeds

technical but unclear

Reality Check

perfect but impossible

Takeover

trust ignored

Blowhard

smart but unusable



ENTHUSIAST, CYNIC, OR DATA HEAD?

Extremes

Enthusiast: data solves everything

Cynic: experience beats data

Both miss something

Data Head

Uses data where useful

Questions it carefully

Communicates with empathy

FINAL TAKEAWAY

WHAT BEING A DATA HEAD MEANS

Do not let the data fool you.

Do not let yourself fool you.

Do not let communication destroy the analysis.

Good data work requires skepticism, humility, and clear communication.



THANK YOU

Any questions?

